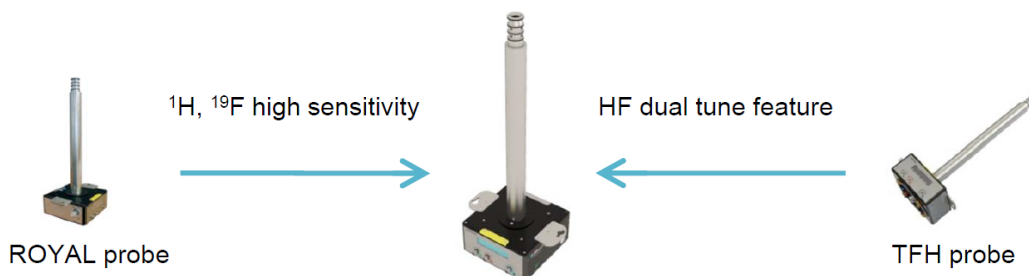
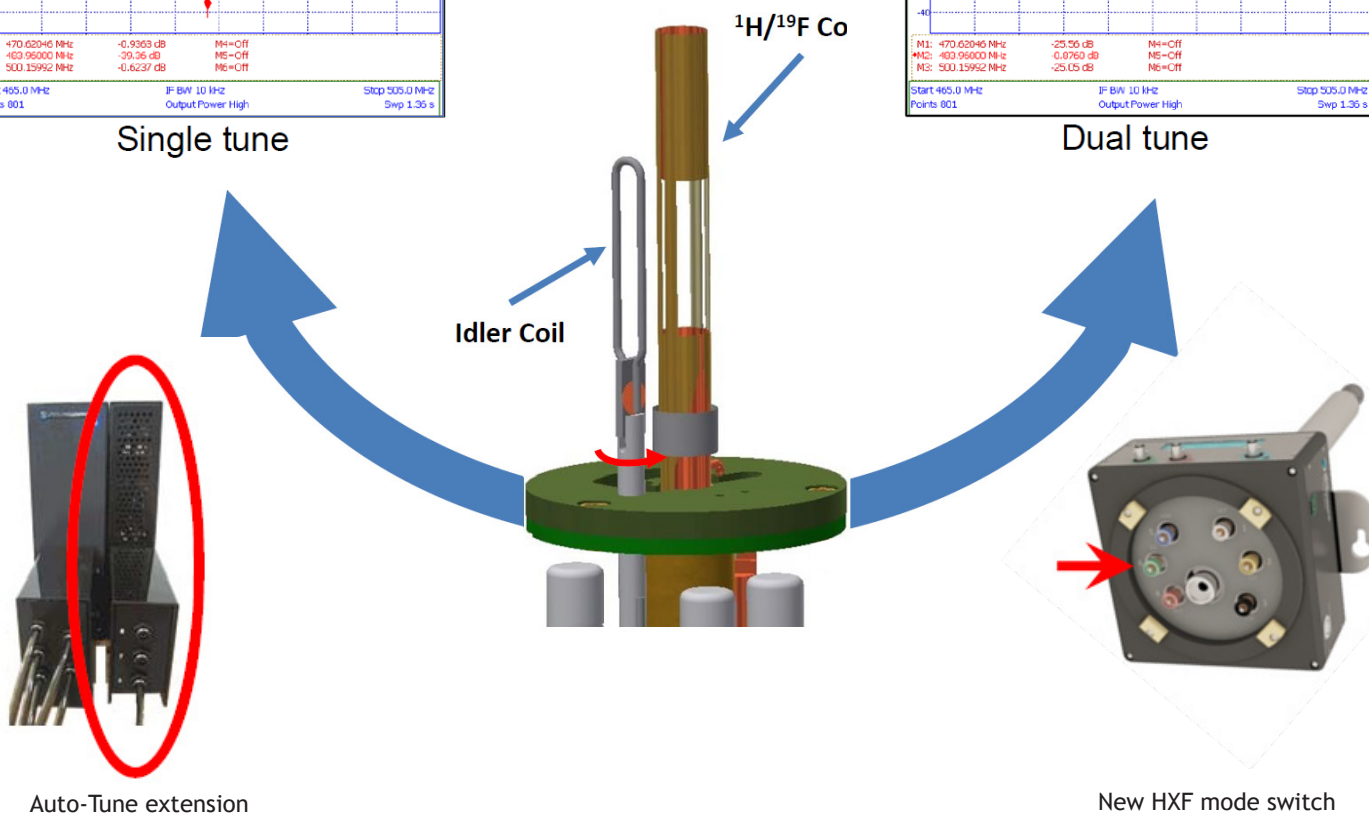
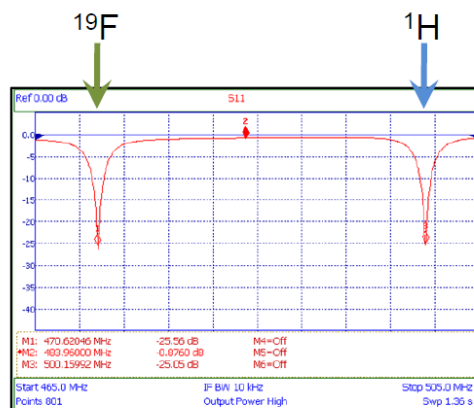
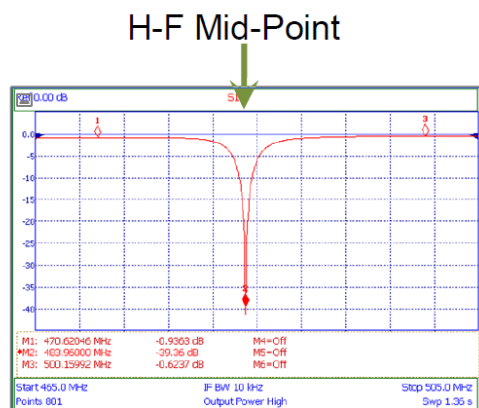


JEOL ROYAL HFX Probe

The JEOL ROYAL HFX NMR probe is the world's first liquid NMR probe switchable between single tune and dual tune mode on the High Frequency coil without compromising the NMR performance. The ROYAL HFX probe operating in single tune mode has the same sensitivity and pulse width performance as the standard ROYAL NMR probe.



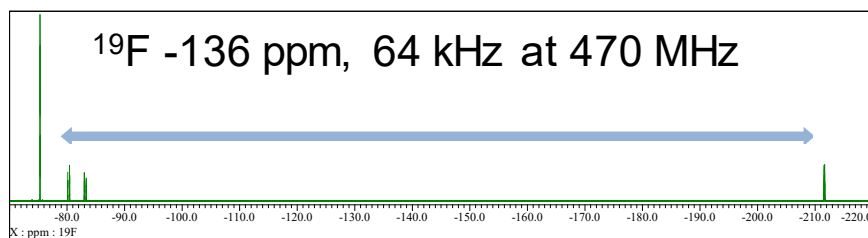
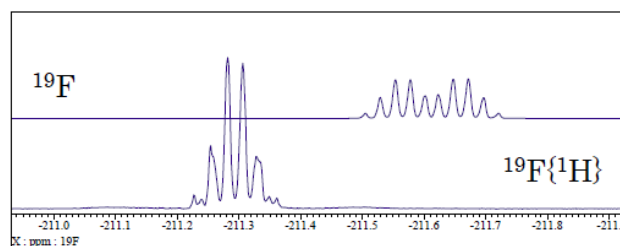
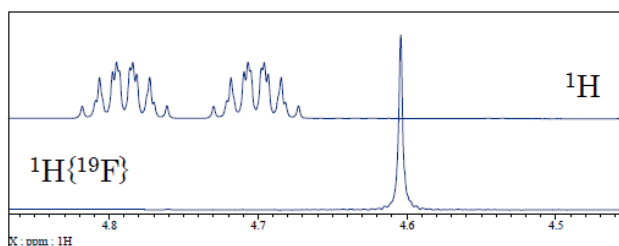
Two probes merged into one.
The JEOL ROYAL HFX NMR Probe



APPLICATION EXAMPLES OF ROYAL HFX PROBE

Operating the HFX probe in dual tune mode allows for a wide variety of ^1H and ^{19}F of advanced NMR experiments including $^1\text{H}\{^{19}\text{F}\}$, $^{19}\text{F}\{^1\text{H}\}$, $^{13}\text{C}\{^1\text{H}, ^{19}\text{F}\}$, and many unique $^{13}\text{C}\{^1\text{H}, ^{19}\text{F}\}$ correlation experiments to simplify spectral assignments of modern complex fluorine containing compounds for the pharmaceutical and polymer industries.

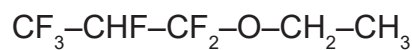
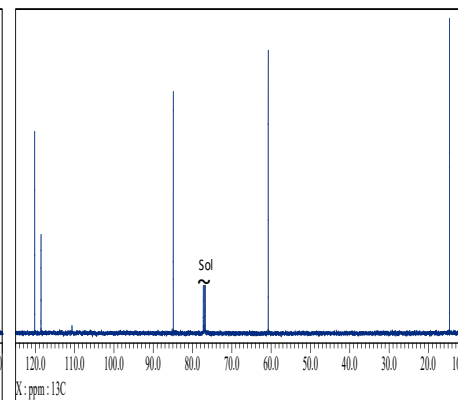
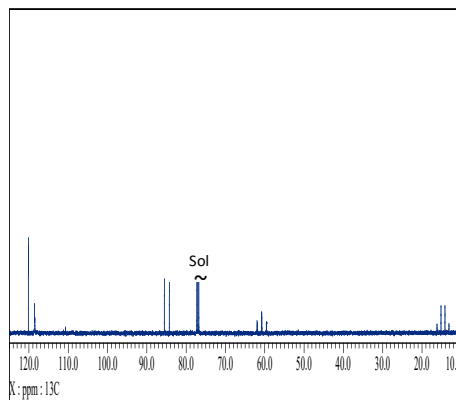
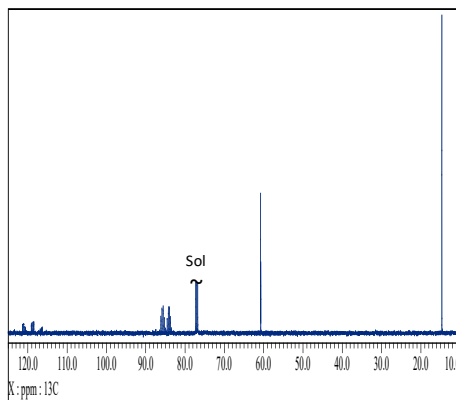
^{19}F -NMR examples measured on a standard two RF channel JNM-ECZ500R



$^{13}\text{C}\{^1\text{H}\}$

$^{13}\text{C}\{^{19}\text{F}\}$

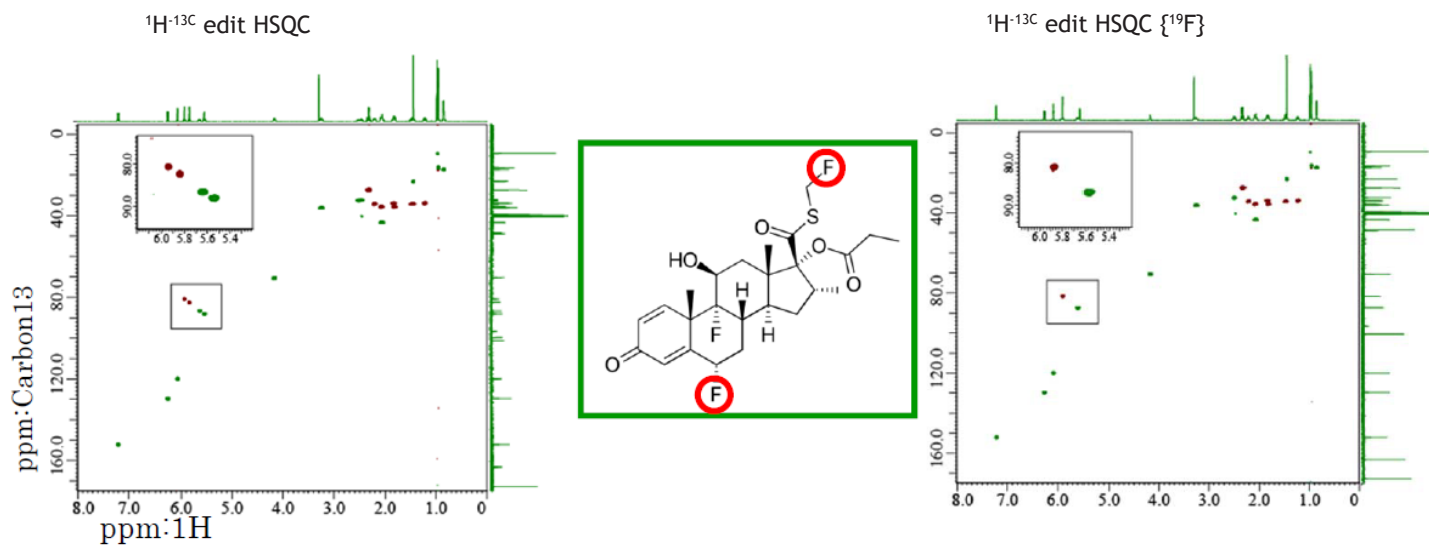
$^{13}\text{C}\{^1\text{H}, ^{19}\text{F}\}$



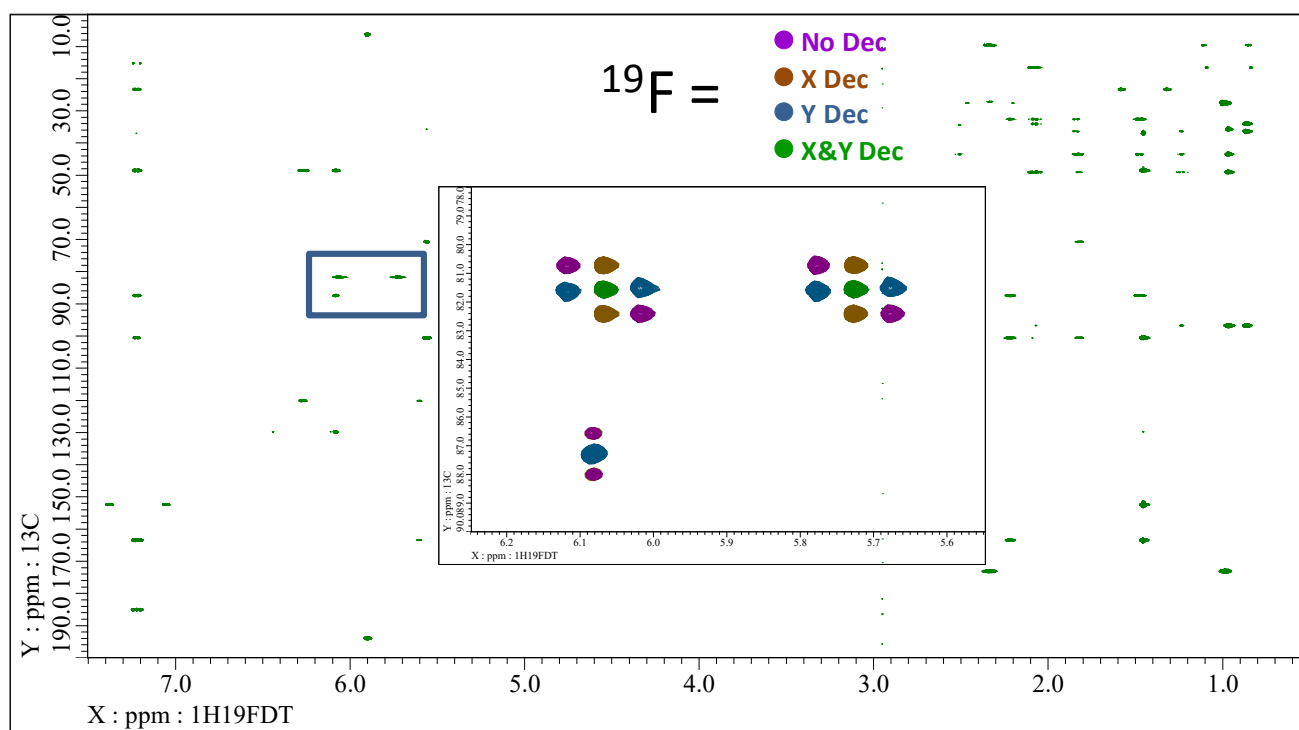
25mg, 1,1,2,3,3,3-Hexafluoropropyl ethyl ether / CDCl_3 ^{13}C -NMR 256 scans

APPLICATION EXAMPLES OF ROYAL HFX PROBE

2D ^1H - ^{13}C HSQC with ^{19}F decoupling, collected on a JEOL JNM-ECZ500R with optional HF2 ^{19}F 3rd channel.



CRISIS-HSQCAD, 25% NUS 4scans

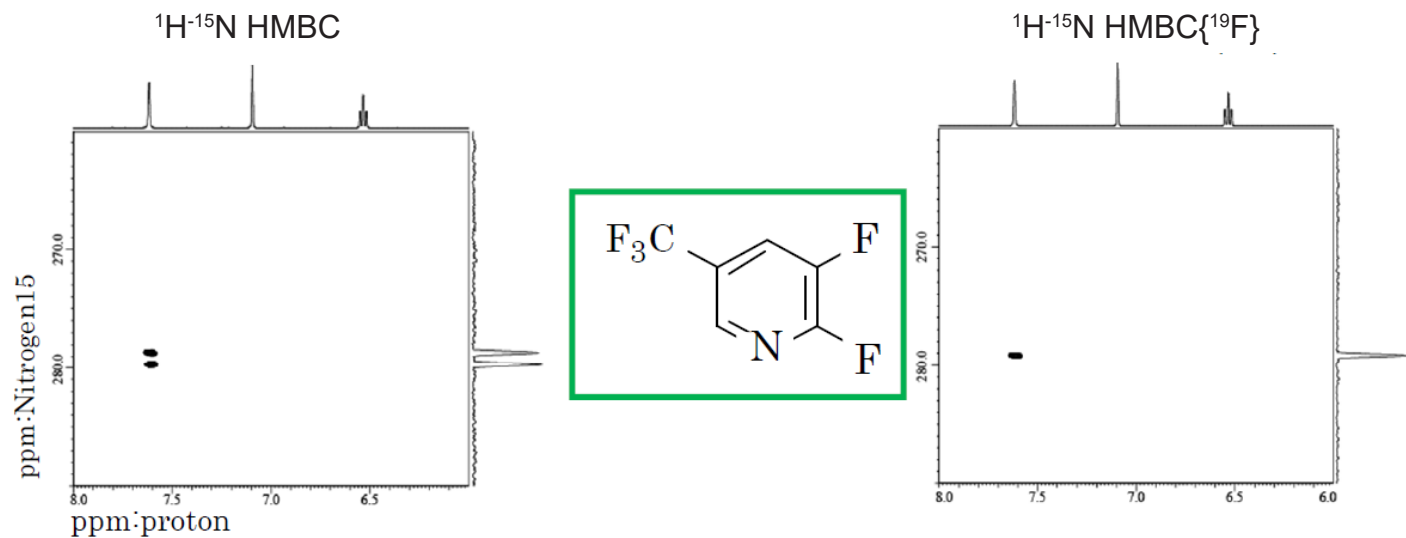
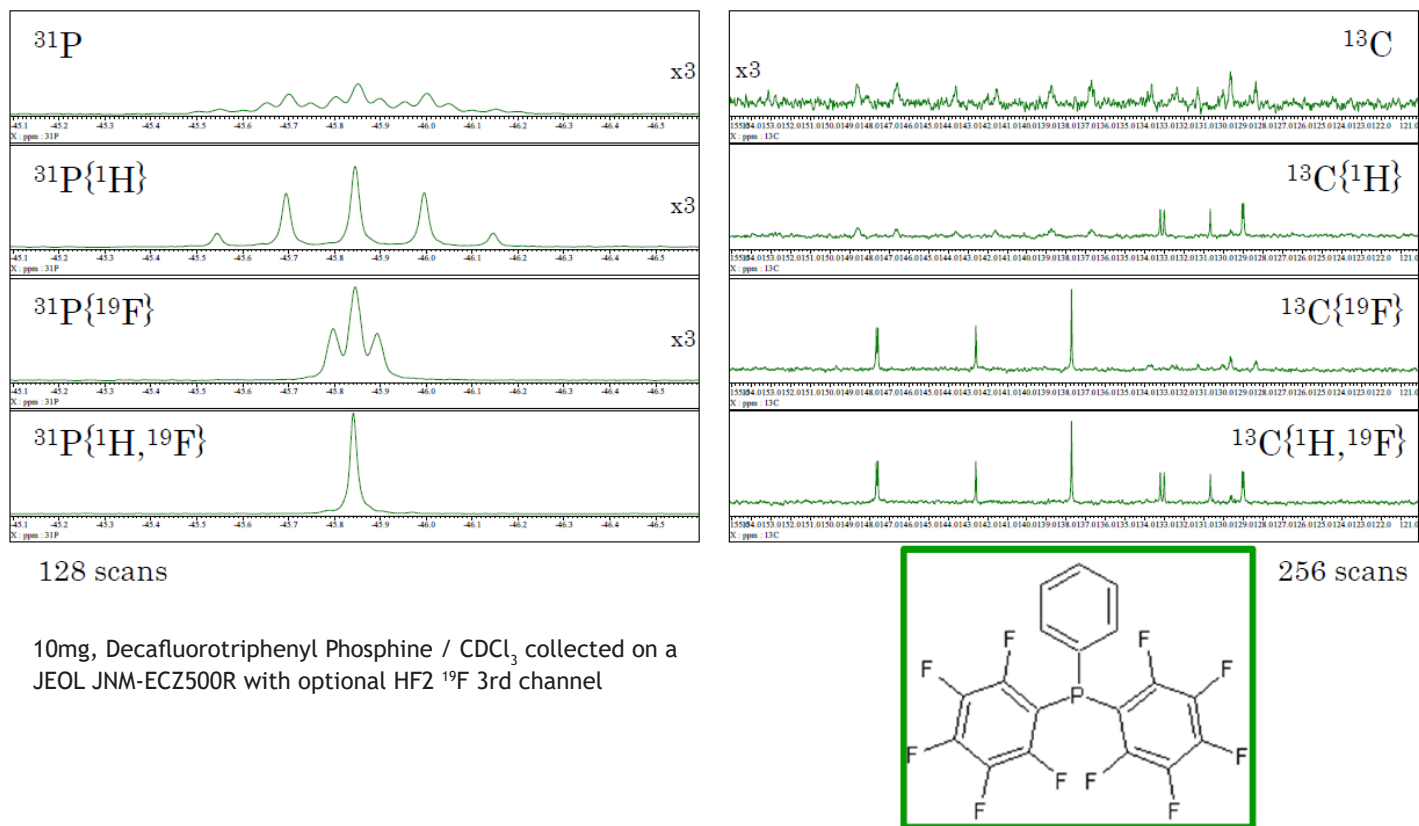


^1H - ^{13}C gHMBCAD, ^{19}F Decoupling Simplifies Assignments, 25% NUS, 8 Scans

10mg Fluticasone Propionate / DMSO-d_6

APPLICATION EXAMPLES OF ROYAL HFX PROBE

The Low Frequency coil is not limited to ^{13}C , but can be tuned to any nucleus between ^{31}P and ^{15}N with the same experimental flexibility as ^{13}C experiments, a feature unique to the ROYAL HFX NMR Probe.



25mg 2,3-difluoro-5-(trifluoromethyl)pyridine / Benzene- d_6 ^1H - ^{15}N HMBC 4scans